

In Angular, **SCSS variables are compiled at build time**, so you **cannot directly change them at runtime**.

If you need runtime styling changes, you must use **CSS custom properties (CSS variables)** instead, because they are resolved by the browser at runtime.

✓ Recommended Approach: Use CSS Variables in Component SCSS

You can still write SCSS, but define **CSS variables** inside it.

Then, you can update them dynamically from your component's TypeScript code.

Example

my-component.component.scss

```
:host {  
  /* Define CSS variables with default values */  
  --primary-color: #3498db;  
  --padding-size: 10px;  
  
  background-color: var(--primary-color);  
  padding: var(--padding-size);  
  color: white;  
  display: block;  
}
```

my-component.component.ts

```
import { Component, ElementRef, Renderer2 } from '@angular/core';

@Component({
  selector: 'app-my-component',
  template: `<p>Dynamic SCSS variables with CSS vars!</p>`,
  styleUrls: ['./my-component.component.scss']
})
export class MyComponent {
  constructor(private el: ElementRef, private renderer: Renderer2) {}

  changeTheme(color: string, padding: string) {
    // Update CSS variables at runtime
    this.renderer.setStyle(this.el.nativeElement, '--primary-color', color);
    this.renderer.setStyle(this.el.nativeElement, '--padding-size', padding);
  }
}
```

Usage in Template

```
<button (click)="changeTheme('#e74c3c', '20px')">Change Theme</button>
```

◆ Why This Works

1. **SCSS variables** (\$var) are replaced at **compile time** → cannot change after build.
2. **CSS variables** (--var) are resolved at **runtime** → can be updated via JavaScript or Angular bindings.

3. You can still use SCSS to organize your styles, but for anything dynamic, wrap it in `var(--name)`.

◆ Alternative: Global Theme Service

If you want **multiple components** to share runtime variables:

1. Define CSS variables in `styles.scss` under `:root`.
2. Update them via a shared Angular service that manipulates `document.documentElement.style`.

If you want, I can give you a **full Angular example with a global theme switcher** so you can change SCSS-like variables across the whole app at runtime.

Do you want me to prepare that?